How Should We Read ViT Attention Maps? A Fast-Weight Perspective

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Abstract

What does a bright patch in a ViT attention map explain? From a fast-weight perspective, we test three assumptions with matched controls: image specificity, spatial correspondence and output contribution. Relocating deep scores by decoded position reduced alignment in three checkpoints. Weighting attention by transformed-value length improved foreground and human-map alignment in supervised DeiT. On Oxford-IIIT Pet, shuffling lengths across source slots reduced foreground enrichment in DeiT-S. In a 128-image subset, using only the final classification query better ranked the measured effects on class margin, while pooling all queries better localised foreground.

1 Introduction

A background patch can attract strong attention in a vision transformer [11]. What does the bright spot say about that part of the image? A received-attention map sums the weights assigned to a source token and displays the sum at its grid position. In deep layers, the token has exchanged information with other positions. The fast-weight view exposes another distinction: the coefficient is the query's activation of that source, while the transformed value determines its vector contribution.

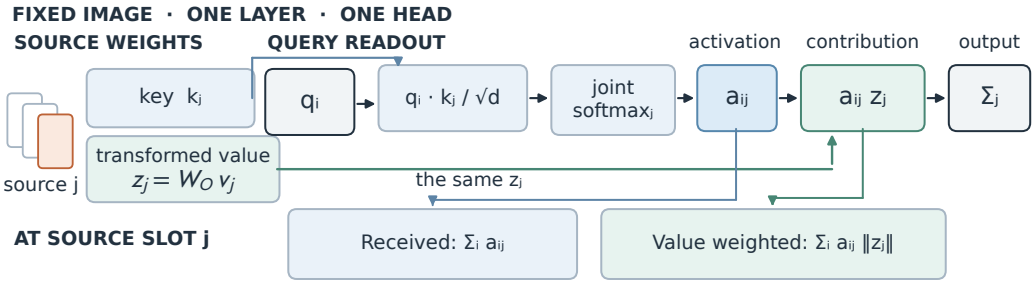


Figure 1: One attention head, read as an image-conditioned two-layer network. For this image, source j supplies input weight $k_j/\sqrt{d}$ and output weight $z_j = W_O v_j$. Softmax over all sources gives activation a_{ij}. Multiplication by z_j gives the contribution to query i. Spatial maps sum a_{ij} or $a_{ij}\|z_j\|_2$ across selected queries and heads at slot j. Head superscripts, output bias and residual are omitted (Eq. 1).

We use MiTA's fast-weight representation [31] to test how attention becomes a spatial explanation:

(1) We pair each reading assumption with a diagnostic, a change in readout and a matched control.
(2) Relocating deep scores by token index reduces alignment across three checkpoints.
(3) We vary value-length placement and query pooling separately to test what a foreground map and a ranking of classifier effects capture.

Preprint.

2 Related work

From attention to a readable unit. Vision transformers mix patch tokens by attention [13]. Fast-weight memory lets a network write temporary weights [28]. Linear attention admits such an interpretation [27], and softmax attention has connections to Hopfield memory [26]. We use MiTA's two-layer representation of softmax attention [31] to identify choices made when displaying a frozen model's activations.

From a token to an image region. Contextualised language tokens retain information about their identity [7], and ViT representations retain spatial information [25]. Registers alter high-norm tokens [11, 18]; no-op heads offer another account of outliers [6]. Rollout, relevance propagation and gradient weighting trace information across layers [1, 9, 10]. Norm analyses incorporate transformed values [19, 36]. GlobEnc and ALTI track layerwise transformations [15, 22], while TokenTM follows token lengths and directions across layers [34]. We ask whether a contextualised token's displayed slot corresponds to image evidence. Centre bias in CNNs was largely absent in the ViTs tested by Kuang et al. [20]; we measure each checkpoint's recurring structure.

Which output is explained? Attention's explanatory value depends on the test [17, 32]. CAM targets a class [35]; Value Zeroing measures contextual mixing in language encoders [23]. Ayyar et al. [3] evaluate filtered attention using human spatial alignment and perturbations. Evaluation also depends on model parameters, supervision and occlusion [2, 5, 30]. We fix image, layer and value weighting, then vary only the receiving queries in patch rankings. We compare those rankings with foreground masks and a direct intervention on the classification-token (CLS) pathway.

3 Methods for reading a frozen attention head

For query i, source j and head h, let $z_j^h = W_O^h v_j^h$ be the transformed value. For query and key dimension d and affine projections, the output before residual addition is

$$o_i = b_O + \sum_{h,j} a_{ij}^h z_j^h, \qquad a_{ij}^h = \mathrm{softmax}_j(q_i^{h\top} k_j^h / \sqrt{d}). \tag{1}$$

For a fixed image, $k_j^h/\sqrt{d}$ and z_j^h act as the input and output weights of a two-layer network in q_i^h, with the softmax coupling its activations across all sources. Received attention (R0) assigns $s_j = \sum_{h,i} a_{ij}^h$ to patch slot j. The sum includes all receiving queries, including CLS, and reports 196 patch scores. Following work that measures properties of hidden units [4], we pair a diagnostic with a controlled change in readout for each assumption.

Image specificity. A bright slot may reflect a recurring spatial pattern. We estimate how much spatial variance a held-out mean map explains and subtract that map from each image's scores (mean subtraction, C1). Subtracting a centred Gaussian scaled to the mean map's spatial standard deviation controls for a radial prior.

Spatial correspondence. A patch slot contains contextual information. We train a linear probe to decode each token's grid index and redistribute s_j by its predicted position probabilities (probe relocation, C2). A blur matched in mean squared displacement tests whether redistribution adds anything beyond smoothing. Index accuracy measures recoverable position information. The input origin of a token's content remains unknown.

Output contribution. An activation alone omits the vector it multiplies. The diagnostic measures the share of received attention on patches whose *block-input norm* exceeds three times the image median. Its matched control zeros R0 at those patches. Value weighting (C3) scales every coefficient by its *projected-value norm*, $c_j = \sum_{h,i} a_{ij}^h \|z_j^h\|_2$, with no threshold. For one query–source pair,

$$\Big\| \sum_h a_{ij}^h z_j^h \Big\|_2 \le \sum_h a_{ij}^h \|z_j^h\|_2.$$

The left side retains cancellation between heads; C3 sums the right side over queries. These terms describe the attention branch before residuals, MLPs and subsequent token mixing.

4 Experiments on three reading assumptions

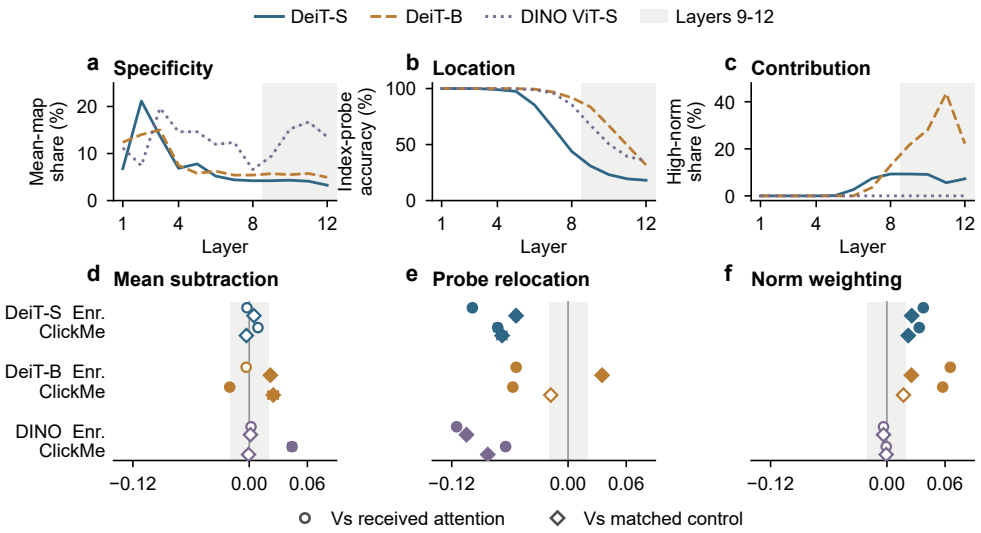


Figure 2: Each column pairs a diagnostic (a–c) with its readout change (C1–C3, d–f). Top: pooled spatial variance explained by the held-out mean, index-probe accuracy and high-norm attention share. Curves have 95% intervals on ImageNet-S (a,c) or 2,000 held-out images (b); shading marks layers 9–12. Bottom: changes from R0 (circles) and matched controls (diamonds) on alignment metrics, with unadjusted paired image-bootstrap 95% intervals. Filled markers exclude zero and reach the 0.02 absolute mean margin. Diagnostic–benefit links are post hoc.

Design and decision rule. We evaluated frozen DeiT-S/B [29] and DINO ViT-S/16 [8], summing layers 9–12. ImageNet-S [16] same-ring enrichment compares top-ranked foreground rate with the rate at the same radii (12,150 valid images per DeiT; 12,144 for DINO). ClickMe [14, 21] alignment is own-minus-other-image Spearman correlation with human maps (17,408 images). Deletion compares top-1 retention when low-ranked versus high-ranked patches are removed first (2,000 images). A higher or lower verdict needs a paired 95% bootstrap interval excluding zero and an absolute mean difference of at least 0.02 (0.05 for deletion); otherwise it is comparable. Primary contrasts compare changes with matched controls and the combination with R0. Individual R0 contrasts are descriptive except R1's registered negative controls. Appendix A gives protocol details.

Mean subtraction (C1). The deep mean map was mostly radial (App. B). Mean subtraction's only gain over R0, on DINO ClickMe, was comparable to a matched Gaussian. In DeiT-B it beat that control on all three metrics but gave no gain over R0.

Probe relocation (C2). Index decoding weakened with depth (Fig. 2b), yet grid index remained decodable. Deep probe relocation lowered all nine model–metric scores relative to R0 and underperformed matched blur in DeiT-S and DINO (Table B.3; App. C). Index decoding does not establish content origin.

Value weighting (C3). Late attention to high-norm patches was concentrated in the supervised models (Fig. 2c). DeiT-S/B value weighting improved enrichment by 0.038/0.065 and ClickMe alignment by 0.033/0.058 over R0. Three of four matched-control comparisons met the registered criterion for improvement. On DeiT-B ClickMe, C3 was 0.017 higher than zeroing high-norm attention, below the 0.02 margin. Deletion gains of 0.017/0.029 over R0 fell below its 0.05 margin.

Registered predictions. R1 held in all nine tests: deep DINO C3 was comparable to R0 on three metrics; early C2 (layers 1–4) was comparable on both alignment metrics in all models. R2 *partly holds*: the combination gave one higher, one comparable and two lower supervised alignment scores. R3 held: neither the test-time register nor MiTA was rated higher than DeiT-S on either alignment metric. MiTA differs in training and preprocessing (App. A).

5 Two exploratory tests of what a readout explains

Which source gets which length? Weighting attention by projected-value length improved foreground and ClickMe alignment in supervised DeiT. On 384 hash-selected Oxford-IIIT Pet test images [24], we kept each recipient's attention and value lengths, but reassigned its sorted lengths to slots ranked by a paired donor within each layer and head. A within-ring shuffle retained radial structure. A norm-only control replaced each head's received attention with its spatial mean.

Table 1: Spatial assignment of value lengths in DeiT-S (layers 9–12). All rows use the same 383 pet crops with foreground and background. The difference column gives reading minus C3 for same-ring enrichment, with paired 95% image-bootstrap intervals. AUROC ranks patches across radii.

Reading	Enrichment	Difference from C3 [95% CI]	AUROC
Received attention (R0)	0.141	−0.019 [−0.023, −0.015]	0.788
Projected value norm (C3)	0.160	—	0.817
Donor-image positions	0.137	−0.022 [−0.026, −0.019]	0.787
Within-ring permutation	0.138	−0.021 [−0.025, −0.018]	0.793
Value norm only	0.120	−0.040 [−0.048, −0.032]	0.757
Unprojected value norm	0.163	0.004 [0.003, 0.005]	0.822
Norm after summing heads	0.156	−0.003 [−0.005, −0.002]	0.809

Donor ranking and within-ring shuffling reduced enrichment by 0.022 and 0.021 (Table 1); donor-pair resampling gave a C3 advantage of 0.022 [0.019, 0.026]. The norm-only control was lower too. Thus both value-length placement and spatial variation in attention mattered for this readout. Unprojected value lengths gave slightly higher enrichment on these images.

Which receiving query matters? The donor and shuffle controls reassign lengths across source slots j while pooling all receiving queries i. We next compared all queries with CLS in the final block, whose output feeds the classifier. On 128 of the same pet images, we held attention fixed and removed each contextualised patch's value term from CLS, one at a time. After the remaining block and classifier, we measured the absolute change in the original top-two logit margin. Within each image, we correlated each readout with the 196 measured effects using Spearman's rank correlation. CLS-only per-head magnitudes improved correlation by 0.041 [0.026, 0.057]. After observing that result, we evaluated the same layer-12 rankings against the masks. CLS-only reduced enrichment and AUROC (Table 2); the paired AUROC difference had a 95% interval of $[-0.110, -0.074]$.

Table 2: Layer-12 readouts on the same 128 pet images. Entries average per-image scores. The final row changes only the receiving queries in the sum of per-head magnitudes. Margin Spearman uses the effect of removing each value term from CLS; net magnitude allows head-vector cancellation.

Layer-12 readout	Margin Spearman	Enrichment	AUROC
All queries, attention	0.696	0.117	0.754
All queries, sum of magnitudes	0.705	0.127	0.769
CLS, attention	0.730	0.081	0.666
CLS, sum of magnitudes	0.747	0.095	0.678
CLS, net magnitude	0.756	0.098	0.678
CLS − all queries	**+0.041**	**−0.033**	**−0.091**

6 Discussion and conclusion

On the same pet images, pooling all queries better localised foreground, while CLS-only rankings tracked final-layer CLS value-term removals more closely. Reading an attention map therefore requires specifying the displayed source slot, pooled receiving queries and outcome to explain. Index decodability did not justify relocating deep scores, whereas value weighting improved alignment in supervised DeiT. The CLS result covers one checkpoint, image domain and final-layer path. Human maps and object masks reflect different annotation tasks. Tracing contextualised-token evidence back to input pixels remains open.

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A Experimental details

Models. All models are frozen ViTs with 16×16 patches on 224×224 inputs. The three baseline backbones come from timm [33]. *DeiT-S and DeiT-B* [29]: `deit_small_patch16_224.fb_in1k` and `deit_base_patch16_224.fb_in1k` (resize to 248, centre crop 224: a crop ratio of 0.9). *DINO ViT-S/16* [8]: `vit_small_patch16_224.dino` with the official DINO linear head (resize to 256, centre crop 224: a crop ratio of 0.875). *DeiT-S with a test-time register* [18]: the activations of eight feed-forward units in layers 4 and 5 are moved to one added register token, which is left out of every map, like CLS. The units were selected on 500 ImageNet-S images, where they reduce the number of high-norm patch tokens (counted per layer) by 92.75% and keep 99.0% of the predictions; on the other 11,919 images the reduction is 91.5%, and ImageNet accuracy stays at 79.8%. For DeiT-B the same search found no such units (44.9% reduction with 32 units), so we report the register for DeiT-S only. *Top-k training*: the DeiT-S checkpoint released with MiTA [31] (25 summary slots, $k = 25$), with its own preprocessing (crop ratio 0.875). It differs from our DeiT-S in training and preprocessing, so its row in Table B.1 is not a controlled comparison; its deletion score is undefined because its summary slots pool fixed regions of the full grid. Both model-side changes are tested at size S only; in a preliminary comparison with an earlier protocol, top-k training scored 0.011 lower in object enrichment than its baseline at size S but 0.037 higher at size Tiny, so the result should not be read beyond DeiT-S.

Attention conventions. With x the output of the block's first layer norm, projections include the checkpoint's biases:

$$q_i^h = W_Q^h x_i + b_Q^h, \quad k_j^h = W_K^h x_j + b_K^h, \quad v_j^h = W_V^h x_j + b_V^h.$$

Equation 1 includes the output bias b_O and describes the attention branch before residual addition, with dropout inactive at evaluation. W_O^h is the head's slice of the output projection. C3 assigns the per-head vector magnitudes to patches; the shared output bias is not assigned to any patch. The fixed-image two-layer representation uses the coupled softmax activation, rather than an elementwise nonlinearity.

Data. *Execution test*: 5,000 randomly drawn ImageNet validation images [12]. *ImageNet-S* [16]: all 12,419 validation images with segmentation masks; stored enrichment is finite for 12,150 (12,144 for DINO). Historical records include all-foreground grids scored zero. Appendix E gives the cohort sensitivity check. Deletion uses a fixed random subset of 2,000. *ClickMe* [14, 21]: the 17,408-image validation split released with Harmonization (17 `val-*.tfrecords` shards). This split name does not identify the original ImageNet split of each image. Images are resized from 256×256 to 224×224 without cropping; heatmaps are resized to 64×64, blurred with a 10×10 Gaussian ($\sigma = 10$), then resized to 224×224 (bilinear interpolation). *Held-out set*: 4,000 ImageNet validation images excluded from map-quality and execution evaluation; 2,000 train probes and 2,000 provide mean maps, probe-test accuracies and blur calibration. Our near-duplicate check matched 2 of 4,002 candidate ImageNet validation images to ClickMe records; this does not establish every ClickMe image's training-set membership. Possible overlap with model training images limits generalisation claims.

Diagnostics. All three are computed per layer. *Specificity*: with $\bar{s}$ the mean received-attention map of the held-out images and Var_j the variance over the 196 patches of one image, the share explained is $1 - \sum_x \mathrm{Var}_j(s - \bar{s}) \,/\, \sum_x \mathrm{Var}_j(s)$, summed over the ImageNet-S images x. *Location*: the top-1 accuracy of the linear position probe on the 196 patch tokens of each of 2,000 held-out images. *Contribution*: the share of received attention on high-norm tokens, on the ImageNet-S images. The high-norm threshold (three times the median, defined below) is an operational definition fixed before the experiments; we did not test other thresholds. The correction itself uses no threshold.

Corrections and matched controls. Scores are computed per layer and summed over layers 9–12 without per-layer rescaling, except for the C1 control's calibration described below. Queries include all 197 tokens; maps contain 196 patches. Negative C1 scores are retained for ranking, without clipping. *Mean-map subtraction* computes $\sum_\ell (s_\ell - \bar{s}_\ell)$ using the 2,000 calibration images. With $S = \sum_\ell s_\ell$ and $\bar{S} = \sum_\ell \bar{s}_\ell$, its control is $S - G\ \mathrm{sd}_j(\bar{S}) / \mathrm{sd}_j(G)$, where G is a centred isotropic Gaussian ($\sigma = 3.5$ patches). Calibration thus uses the *summed* mean map. *Probe relocation* computes $r_p = \sum_j s_j P_\ell(p \mid x_j)$, preserving total score. One linear 196-way classifier per model/layer is

trained on standardised tokens after the first layer norm: cross-entropy, Adam, learning rate 0.01, 30 epochs, token minibatches of 4,096, seed 0. Feature means/standard deviations use only the 2,000 training images; standard deviations are floored at 10^{-6}. Its Gaussian control is a row-normalised kernel $K_{jp} \propto \exp(-\|p-j\|^2/(2\sigma_\ell^2))$. The width (0.7–2.9 patches) matches

$$D_\ell = \frac{1}{2000N} \sum_{x,j,p} P_\ell(p \mid x_j)\|p-j\|^2$$

on calibration images, averaging uniformly over source patch indices. ClickMe uses the same crop-trained probe and blur on resized full images, introducing a potential geometry mismatch. *Contribution weighting* computes $c_j = \sum_{h,i} a_{ij}^h \|W_O^h v_j^h\|_2$, a sum of per-head norm-weighted terms related to Kobayashi et al. [19]. Summing magnitudes before aggregating across heads omits cross-head cancellation; residual addition, subsequent normalisation and later-layer computation are outside this score. The control zeros received attention at high-norm patches: the residual-stream block-input norm *before* the first layer norm must exceed three times the image's median patch norm, using the midpoint median. C3 itself has no high-norm threshold. *All three corrections*: contribution weighting, probe relocation, then subtraction of the held-out mean of the resulting map, per layer before summation.

Metrics. *Same-ring object enrichment.* A patch is valid if it contains at least one non-ignored pixel, and is an object patch if at least half of its non-ignored pixels belong to the image's class mask. We select as many top-scoring patches as there are object patches. For every ring of patches at the same distance from the grid centre, we take the object rate among its selected patches minus that among all its patches, and average over rings, weighted by their selected patches. Invalid patches are excluded from selection and ring baselines. Ties are broken with the fixed seeded perturbation used in the original runs. A purely radial map scores zero in expectation over tie-breaking. *Image-specific ClickMe alignment.* The map is upsampled to 224×224 and correlated (Spearman, over pixels) with the image's own human map; we subtract its correlation with the previous image's human map within each batch (`roll(1)`; batch size 64). For an image-independent map, these differences average to zero over the paired set because the control permutes the same human maps; individual-image differences need not be zero. This removes fixed-prior alignment on average, without guaranteeing invariance to a prior mixed into an image-dependent map. *Deletion score.* Patches are ranked once by the original map. At removal budgets $0, 1, 2, 4, 8, 16, 32, 64, 98, 147$, we record the binary indicator that the original top-1 prediction is preserved. Tokens are dropped after positional embedding; CLS and any register remain. We use an SRG-style LIF–MIF area difference [5], adapting the value function from class probability to top-1 preservation. Areas use trapezoidal integration over removed-patch fraction $b/196$, from 0 to $147/196$. *Validity of the metrics.* A random map, a centred Gaussian and the mean map all score within 0.004 of zero on enrichment and on ClickMe alignment (the two image-independent maps have exactly zero mean on the latter, up to numerical precision); on deletion the centred Gaussian scores 0.110, 0.099 and 0.140 (DeiT-S, DeiT-B, DINO ViT-S), which is why every correction is compared with a matched control. *Statistics.* One value per image; paired image-bootstrap with 1,000 resamples, seed 0 and 95% percentile intervals. A difference is *higher* (*h*) or *lower* (*l*) if its interval excludes zero and its absolute mean reaches 0.02 (0.05 for deletion), and *comparable* (*c*) otherwise. The comparable category also covers imprecise differences. Intervals are unadjusted for multiple comparisons. Resampling holds checkpoints, probes and the within-batch ClickMe pairing fixed, excluding uncertainty from training, probe fitting or alternative pairings. Because the control reuses human maps within each batch, ClickMe score rows are not independent pairings.

GAE reference. The gradient-based reference [10] targets the unmodified model's predicted top-1 logit and uses all 12 attention layers. Starting with $R = I$, each layer updates $R \leftarrow R + \mathrm{mean}_h[(\nabla_{A^h} f_c \odot A^h)_+]R$, then takes the final CLS-to-patch row. It is class-specific and spans all layers, whereas R0/C1–C3 are class-agnostic and sum layers 9–12. Its scores provide descriptive context rather than a matched single-component ablation.

Registered predictions. The protocol was frozen before formal runs, after smoke tests on 32–64 images and earlier exploratory observations documented in `docs/preregistration.md`. Historical P4/P3/P6 map to R1/R2/R3; definitions, margins and verdicts are unchanged. R2 has three possible outcomes: holds (4/4 higher), partly holds (1–3) and does not hold (0). Primary contrasts

were the combination versus R0 and each correction versus its matched control, without a registered direction for the latter. Other correction–R0 contrasts are descriptive, except R1's registered negative controls. Prior execution/diagnostic results had been seen, making the last two rows of Table A.1 replications. Diagnosis–benefit interpretation and supplementary mechanism analyses are post hoc.

Table A.1: Registered predictions and outcomes (centre-controlled metrics: enrichment and ClickMe alignment).

	Prediction	Outcome
R1	Contribution weighting is comparable to received attention in DINO ViT-S on all three metrics	holds (3 of 3 cells)
R1	Relocation at layers 1–4 is comparable to received attention on the centre-controlled metrics in all models	holds (6 of 6 cells)
R2	All three corrections together are higher than received attention on the centre-controlled metrics in DeiT-S and DeiT-B	partly holds (1 higher: DeiT-B ClickMe; 1 comparable; 2 lower)
R3	Neither model-side change is higher than DeiT-S received attention on the centre-controlled metrics	holds (4 of 4 cells)
–	Execution agreement rises with k and exceeds that of random sets at every k	holds in all models
–	Probe accuracy falls with depth; the share of attention on high-norm tokens peaks in layers 6–12 of the supervised models and is below 1% in every layer of DINO ViT-S	holds in all models

Compute and code. The original experiments used approximately 19 NVIDIA L4 GPU hours. The paper repository provides the registered protocol, configurations and fixed splits in `docs/preregistration.md`, `configs/` and `data/splits/`. The exploratory protocols, scripts and individual scores are in `experiments/`. The README gives separate commands for rebuilding the displayed results and running inference.

B Additional results

These results support the framework at different levels: execution and parameter sensitivity check internal dependence, while paired comparisons and supplementary analyses assess spatial alignment, deletion sensitivity and proposed interpretations.

Table B.1: Complete scores for the original tests, layers 9–12 unless indicated. Arrows mark higher/lower scores versus received attention (DeiT-S for model-side rows). Diamonds indicate higher scores than the matched control. Bold indicates higher than both. Untagged tested cells are comparable. GAE uses gradients and all 12 layers. Dashes indicate unavailable comparisons.

	DeiT-S			DeiT-B			DINO ViT-S		
Reading	Enr.	Click.	Del.	Enr.	Click.	Del.	Enr.	Click.	Del.
Received attention	0.140	0.157	0.236	0.080	0.107	0.169	0.199	0.208	0.285
Contribution weighting	**0.178**$^{\uparrow\diamond}$	**0.190**$^{\uparrow\diamond}$	0.253	**0.146**$^{\uparrow\diamond}$	0.165$^{\uparrow}$	0.198	0.196	0.207	0.285
zero high-norm tokens	0.152	0.168	0.241	0.121$^{\uparrow}$	0.148$^{\uparrow}$	0.186	0.199	0.208	0.285
Mean-map subtraction	0.138	0.165	0.208	0.077$^{\diamond}$	0.087$^{\downarrow\diamond}$	0.144$^{\diamond}$	0.201	0.252$^{\uparrow}$	0.233$^{\downarrow}$
subtract centre Gaussian	0.133	0.168	0.167$^{\downarrow}$	0.056$^{\downarrow}$	0.062$^{\downarrow}$	0.024$^{\downarrow}$	0.200	0.253$^{\uparrow}$	0.232$^{\downarrow}$
Probe relocation	0.041$^{\downarrow}$	0.084$^{\downarrow}$	0.076$^{\downarrow}$	0.027$^{\downarrow\diamond}$	0.050$^{\downarrow}$	0.051$^{\downarrow}$	0.084$^{\downarrow}$	0.143$^{\downarrow}$	0.141$^{\downarrow}$
matched blur	0.095$^{\downarrow}$	0.152	0.181$^{\downarrow}$	−0.008$^{\downarrow}$	0.068$^{\downarrow}$	0.097$^{\downarrow}$	0.189	0.226	0.287
All three corrections	0.059$^{\downarrow}$	0.123$^{\downarrow}$	0.076$^{\downarrow}$	0.079	0.137$^{\uparrow}$	0.085$^{\downarrow}$	0.088$^{\downarrow}$	0.177$^{\downarrow}$	0.090$^{\downarrow}$
+ test-time register	0.152	0.169	0.236	–	–	–	–	–	–
Top-k training (MiTA)	0.100$^{\downarrow}$	0.115$^{\downarrow}$	–	–	–	–	–	–	–
GAE (reference)	0.204	0.234	0.345	0.196	0.213	0.308	0.221	0.234	0.418

Execution test. The 14×14 grid is pooled into 16 overlapping regions. In every layer/head, each region's mean query selects its top-k keys; CLS retains its own set. Sets from an unmodified forward pass restrict every query to its region's set in all 12 layers simultaneously. Top-k sets beat equal-size random sets at every tested k (4–128) in all models, and agreement rises with k (Fig. B.1). The dotted attention-removed baseline alone uses uniform crop ratio 0.875; solid/dashed DeiT curves use 0.9.

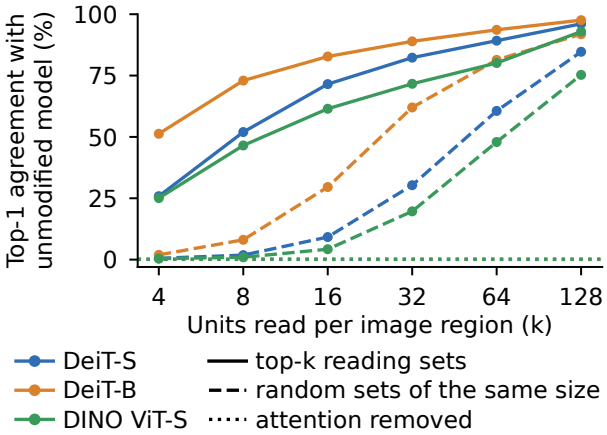


Figure B.1: Top-1 agreement on 5,000 ImageNet images with region-selected top-k units (solid), random units (dashed) or attention removed (dotted). Stored 95% intervals are at most ±1.5 percentage points. Dotted curves use crop ratio 0.875 for all models; other curves use each checkpoint's main preprocessing.

Diagnostics per layer. Table B.2 lists the diagnostics plotted in Fig. 2a–c and the uncorrected per-layer ClickMe alignment. Two further details complement the table. First, in layers 9–12, 82–98% of the variance of the mean map is explained by the distance from the grid centre. Second, averaged over layers 9–12, the mean value-vector norm $\|v_j\|$ of high-norm tokens is 0.29 (DeiT-S) and 0.23 (DeiT-B) times the median of the other patches (measured on 11,919 ImageNet-S images using uniform crop ratio 0.875, unlike the main DeiT evaluation at 0.9).

Recovery at the last layer. In the supervised models, single-layer received attention is worse at layer 12 than at the best of layers 8–11 (ClickMe values in Table B.2). Applied to layer 12, contribution weighting recovers 62% and 33% of this drop on enrichment (DeiT-S, DeiT-B), 31% and 45% on ClickMe alignment, and 8% and 29% on deletion. The other two corrections make layer 12 worse.

Differences behind Table B.1. Table B.3 gives the paired differences with their intervals. The values in Table B.1 have 95% intervals of at most ±0.005 on enrichment and ClickMe alignment and ±0.012 on deletion. The model-side differences from DeiT-S received attention are $+0.012$ [0.011, 0.013] on enrichment, $+0.013$ [0.012, 0.013] on ClickMe and 0.000 [-0.005, 0.004] on deletion for the test-time register, and -0.040 [-0.042, -0.039] and -0.042 [-0.044, -0.040] on the first two for top-k training. The paired MiTA enrichment contrast uses 12,118 images valid for both models.

Table B.2: Per-layer diagnostics (Fig. 2a–c) and uncorrected ClickMe alignment (last three columns). S, B, D: DeiT-S, DeiT-B, DINO ViT-S; probe chance: 0.5%. Values are rounded; DINO layers 9–12 have exactly zero high-norm attention on this dataset.

	Specificity: mean-map share (%)			Location: probe accuracy (%)			Contribution: high-norm share (%)			Map quality: ClickMe alignment		
Layer	S	B	D	S	B	D	S	B	D	S	B	D
1	6.8	12.4	11.2	100.0	100.0	99.9	0.0	0.0	0.0	−0.108	−0.111	−0.089
2	21.2	14.0	7.4	100.0	100.0	100.0	0.0	0.0	0.0	0.007	0.041	0.064
3	13.7	15.0	19.7	99.9	100.0	99.9	0.0	0.0	0.0	0.081	0.004	0.056
4	6.9	7.5	14.6	99.0	100.0	99.8	0.0	0.0	0.0	0.132	0.110	0.140
5	7.8	5.8	14.6	97.5	99.9	99.6	0.1	0.0	0.0	0.148	0.130	0.150
6	5.2	6.2	11.9	85.3	99.5	98.9	2.6	0.0	0.0	0.136	0.151	0.167
7	4.4	5.4	12.4	65.0	96.8	95.9	7.4	3.5	0.0	0.147	0.131	0.150
8	4.2	5.4	6.6	43.9	91.8	85.6	9.3	12.9	0.0	0.149	0.118	0.170
9	4.2	5.7	9.5	31.0	83.6	67.4	9.3	21.5	0.0	0.184	0.112	0.198
10	4.3	5.5	15.2	23.1	66.8	50.5	9.1	28.1	0.0	0.171	0.139	0.207
11	4.1	5.8	16.7	19.4	49.2	39.2	5.6	43.5	0.0	0.149	0.120	0.211
12	3.3	5.0	13.6	18.1	31.8	35.7	7.2	22.2	0.0	0.098	0.023	0.179

Table B.3: Paired differences, layers 9–12: mean, 95% image-bootstrap interval and verdict (*h*/*c*/*l*: higher/comparable/lower). Margins: 0.02 (enrichment/ClickMe), 0.05 (deletion); intervals unadjusted. Samples: App. A. Top: descriptive correction–R0 contrasts and the primary combination–R0 contrasts. Bottom: primary correction–control contrasts. Registered negative controls are also discussed in App. A.

Model	Same-ring enrichment	ClickMe alignment	Deletion score
Contribution weighting − received attention			
DeiT-S	0.038 [0.036, 0.039] *h*	0.033 [0.032, 0.034] *h*	0.017 [0.013, 0.021] *c*
DeiT-B	0.065 [0.063, 0.067] *h*	0.058 [0.056, 0.059] *h*	0.029 [0.024, 0.034] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − received attention			
DeiT-S	−0.002 [−0.003, −0.002] *c*	0.009 [0.008, 0.010] *c*	−0.028 [−0.034, −0.023] *c*
DeiT-B	−0.003 [−0.004, −0.003] *c*	−0.020 [−0.021, −0.019] *l*	−0.025 [−0.031, −0.019] *c*
DINO ViT-S	0.002 [0.001, 0.003] *c*	0.044 [0.040, 0.048] *h*	−0.052 [−0.059, −0.044] *l*
Probe relocation − received attention			
DeiT-S	−0.099 [−0.101, −0.097] *l*	−0.073 [−0.076, −0.069] *l*	−0.161 [−0.171, −0.150] *l*
DeiT-B	−0.054 [−0.055, −0.052] *l*	−0.057 [−0.059, −0.055] *l*	−0.118 [−0.127, −0.110] *l*
DINO ViT-S	−0.115 [−0.117, −0.113] *l*	−0.064 [−0.067, −0.062] *l*	−0.144 [−0.153, −0.134] *l*
All three corrections − received attention			
DeiT-S	−0.081 [−0.083, −0.079] *l*	−0.034 [−0.038, −0.030] *l*	−0.160 [−0.171, −0.149] *l*
DeiT-B	−0.001 [−0.003, 0.001] *c*	0.030 [0.027, 0.032] *h*	−0.084 [−0.093, −0.076] *l*
DINO ViT-S	−0.111 [−0.114, −0.109] *l*	−0.031 [−0.035, −0.027] *l*	−0.195 [−0.205, −0.184] *l*
Contribution weighting − zero high-norm tokens (its control)			
DeiT-S	0.026 [0.025, 0.027] *h*	0.022 [0.021, 0.023] *h*	0.012 [0.008, 0.016] *c*
DeiT-B	0.025 [0.024, 0.026] *h*	0.017 [0.016, 0.018] *c*	0.013 [0.008, 0.018] *c*
DINO ViT-S	−0.004 [−0.004, −0.003] *c*	−0.001 [−0.001, −0.001] *c*	0.000 [−0.003, 0.004] *c*
Mean-map subtraction − centre-Gaussian subtraction (its control)			
DeiT-S	0.005 [0.004, 0.005] *c*	−0.003 [−0.005, −0.001] *c*	0.040 [0.034, 0.047] *c*
DeiT-B	0.022 [0.021, 0.023] *h*	0.025 [0.020, 0.030] *h*	0.120 [0.109, 0.130] *h*
DINO ViT-S	0.001 [0.001, 0.002] *c*	−0.001 [−0.001, 0.000] *c*	0.001 [−0.003, 0.005] *c*
Probe relocation − matched blur (its control)			
DeiT-S	−0.054 [−0.056, −0.051] *l*	−0.068 [−0.073, −0.064] *l*	−0.105 [−0.116, −0.095] *l*
DeiT-B	0.035 [0.034, 0.036] *h*	−0.018 [−0.020, −0.015] *c*	−0.047 [−0.055, −0.037] *c*
DINO ViT-S	−0.105 [−0.107, −0.102] *l*	−0.083 [−0.085, −0.080] *l*	−0.146 [−0.155, −0.137] *l*

Tests of the contribution account. Norm weighting reduces the relative weight of attention attached to small transformed-value vectors. One consequence was registered (R1); additional checks were post hoc. *No practically meaningful gain in DINO (R1).* In DINO ViT-S, with zero high-norm attention at layers 9–12 on ImageNet-S, every difference from received attention is at most 0.004 in size (Table B.3). *High-norm attention and alignment gain (post hoc).* Across images, gains have a weak positive association with the share of received attention on high-norm tokens: Spearman $+0.075$ [0.057, 0.092] and $+0.132$ [0.113, 0.150] on enrichment (DeiT-S, DeiT-B), $+0.045$ [0.032, 0.060] and $+0.158$ [0.143, 0.173] on ClickMe. *Assigned versus shuffled weights (post hoc).* Against the same weights shuffled among the patches of each head, contribution weighting is higher in all four supervised cells: $+0.038$ [0.037, 0.040] and $+0.073$ [0.071, 0.076] on enrichment, $+0.035$ [0.035, 0.036] and $+0.067$ [0.065, 0.068] on ClickMe. The shuffle control averages seeds 0–4 within each image before bootstrap resampling. Finally, also post hoc, weighting by $\|v_j^h\|$ instead of $\|W_O^h v_j^h\|$ is comparable in every cell (differences -0.010 to 0.000). This comparable verdict does not establish equivalence. C3 aggregates per-head terms related to Kobayashi et al. [19] and shares a norm-weighted component with Zhu et al. [36]; their full analyses and aggregation differ.

B.1 Parameter-randomisation sanity check

The original study included a model-parameter randomisation check [2] on 1,000 fixed-seed ImageNet-S images. Parameters were cumulatively re-initialised from the classifier and final norm through blocks 12–1 to the patch/position embeddings and CLS token. The trained model's probes and mean maps remained fixed. Received attention and its corrections are class-agnostic and therefore do not depend on classifier parameters alone. Table B.4 reports the stored median per-image Spearman correlations between trained and fully randomised maps, with layers 9–12 summed. All five readings meet the registered median < 0.5 criterion; the model-independent centre Gaussian fails as designed. This is a necessary sensitivity check, not proof of prediction faithfulness. The existing summary is reproduced here without rerunning the experiment.

Table B.4: Stored model-randomisation results: median map correlation after full randomisation. The five readings pass the registered < 0.5 criterion in every model.

Reading	DeiT-S	DeiT-B	DINO ViT-S
Received attention	-0.033	0.055	0.071
Mean-map subtraction	0.187	0.366	0.100
Probe relocation	0.063	0.004	-0.002
Contribution weighting	-0.064	0.070	0.114
All three corrections	0.069	0.078	0.025
Centre Gaussian	1.000	1.000	1.000

C Tests of spatial relocation

Relocation lowers every deep-layer cell (Table B.1). Two planned comparisons narrow the problem. Early relocation remains comparable under the registered rule (R1): at layers 1–4 it changes enrichment and ClickMe alignment by at most 0.004 in every model. And in DeiT-S and DINO ViT-S it is lower than a Gaussian matched in mean squared token displacement (Table B.3). The rest of this section is post hoc: Table C.1 lists four explanations that we tested after seeing this result; none meets its specified criterion. Most probability lies near the true patch, and replacing the linear probe with the tested MLP does not restore map quality. This does not exclude all probe-capacity or calibration limitations. The probe targets index decodability rather than input-content provenance; whether this mismatch explains the failed relocation remains untested (Section 4).

Table C.1: Four explanations for probe-relocation failure (post hoc). Model order: DeiT-S, DeiT-B, DINO ViT-S. Cells are the two centre-controlled metrics in the three models; criteria and probe settings are defined in this section.

Explanation	Test (layers 9–12 unless stated)	Result
The probe sends score to distant patches	Share of the probe's error mass more than three patches from the true patch	Small: 2–29% across models and layers. At layer 12, 45%, 66% and 81% of the probability lies in the 3×3 neighbourhood of the true patch
The probe has a fixed position preference	Peak of the probe's mean prediction relative to uniform; its correlation with the mean map	Close to uniform (peak 1.1–1.3 times the uniform value); at most weakly related to the mean map
The linear probe is too weak	Two-layer MLP probe in place of the linear one	Layer-12 accuracy rises by only 1.8, 3.8 and 6.7 points; relocation is still lower than received attention in all six cells
Global tokens are harder to place	Probe accuracy and entropy against a token's similarity to the image's mean token (layers 10–12)	No consistent relation; accuracy falls with globality only in DeiT-B

Probe capacity and explanatory criteria. The tested MLP has a 1,024-unit hidden layer, GELU and a 196-way output. It uses the linear probe's optimiser, epochs, minibatches and standardisation, trained on 1,800 of the original probe-training images with 200 for best-validation-epoch selection; the original 2,000 test images remain separate. After observing the original relocation result, we specified the following criteria before running these follow-up tests: error mass beyond three patches greater than 0.5 at every deep model/layer; fixed preference peak at least twice uniform and mean-map correlation at least 0.5 (with positive lower interval bound) in every deep model/layer; MLP layer-12 accuracy gain above 10 percentage points in some model, with relocation rescued if it is not lower in at least 4/6 alignment comparisons. These tested criteria fail; the finite MLP check cannot exclude all capacity or calibration limitations.

Globality analysis. Excluding high-norm tokens, globality is a token's cosine similarity to the mean of the other retained tokens in the image. Tokens are split into five equal-size rank groups. We relate the group rank to index accuracy and probe entropy, overall and within four equal-count radial groups, at layers 10–12. After observing the original relocation result, we set a follow-up criterion requiring negative accuracy and positive entropy associations (overall intervals and within-ring means) in at least two models. No consistent relationship meets this criterion.

D Example maps

The stored gallery holds 24 ClickMe images drawn at random with a fixed seed. Fig. D.1 shows the two with the smallest ClickMe index, and Fig. D.2 the next four. The image-selection rule is unchanged. Received attention and contribution weighting look much alike, while probe relocation spreads the score over many more patches. The gains in Table B.1 are averages of small changes in how each map ranks its patches, which single maps do not show.

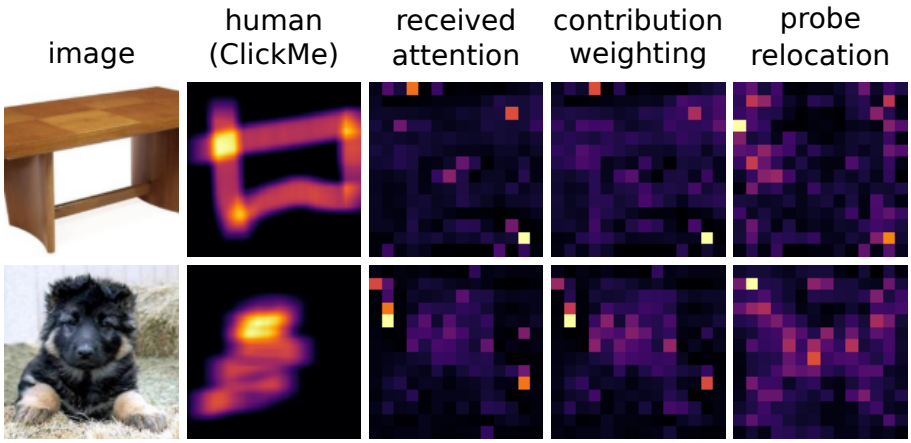


Figure D.1: The first two fixed-seed examples, DeiT-S, layers 9–12. Colours show scores min–max normalised within each map; amplitudes are not comparable across panels. These images were not selected for correction gain.

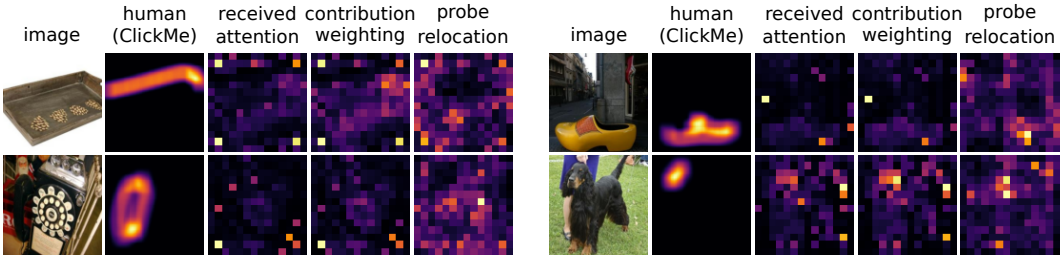


Figure D.2: DeiT-S on four further ClickMe images, layers 9–12; colours are normalised within each map and do not compare amplitudes across panels.

E Exploratory tests of spatial alignment and classifier sensitivity

Deletion endpoints. For each image and method, trapezoidal integration uses the original removal budgets, divided by 196. The probability endpoint integrates the probability of the original top-1 class under least-important-first removal minus most-important-first removal. The KL endpoint uses the reverse difference in divergence from the original prediction, so that larger scores again indicate greater separation. For both endpoints the paired curve difference is zero at budget zero. Results are paired over images. These measures have no registered practical margin.

Uncertainty. Differences are paired by image. Intervals use 1,000 image-bootstrap resamples with seed 0. ClickMe reuses each human map within a batch for its own and another image. Recomputing intervals by sampling the 272 original 64-image batches preserves those dependencies. For C3 relative to zeroing high-norm patches, the resulting ClickMe intervals are [0.021, 0.023] in DeiT-S and [0.016, 0.018] in DeiT-B. All estimates condition on the fixed within-batch pairing and the available checkpoints.

Table E.1: Exploratory endpoint differences (C3 minus comparator). Probability and KL columns give the mean paired area and 95% image-bootstrap interval over the fixed 2,000-image deletion subset. ClickMe uses the original 17,408 images with batch-bootstrap intervals. KL has units of nats times fraction removed.

Model	Comparator	Probability area	KL area	ClickMe
DeiT-S	R0	0.014 [0.012, 0.015]	0.058 [0.050, 0.066]	0.033 [0.032, 0.034]
DeiT-S	Zero high-norm	0.010 [0.009, 0.012]	0.041 [0.033, 0.049]	0.022 [0.021, 0.023]
DeiT-B	R0	0.022 [0.020, 0.024]	0.107 [0.096, 0.118]	0.058 [0.056, 0.059]
DeiT-B	Zero high-norm	0.010 [0.008, 0.012]	0.048 [0.038, 0.058]	0.017 [0.016, 0.018]
DINO	R0	0.000 [−0.001, 0.002]	−0.003 [−0.009, 0.003]	−0.001 [−0.001, −0.001]
DINO	Zero high-norm	0.000 [−0.001, 0.002]	−0.003 [−0.009, 0.003]	−0.001 [−0.001, −0.001]

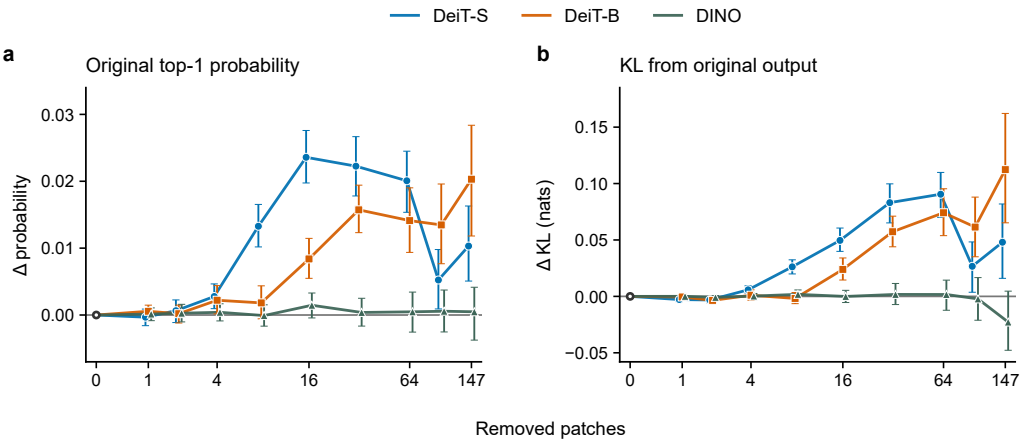


Figure E.1: C3 minus the zero-high-norm control at each original ImageNet-S deletion budget. The probability panel measures least-important-first minus most-important-first target probability. The KL panel reverses the subtraction to express divergence in the same direction. Horizontal positions use a log scale in one plus the number of removed patches, with model-specific offsets for visibility. Intervals are pointwise 95% paired image-bootstrap intervals over 2,000 images per model. The integrated advantage can coexist with a reversal at a small budget.

E.1 Object geometry and the enrichment scale

For an image with M valid patches and $k > 0$ foreground patches, let ring r contain m_r valid patches of which f_r are foreground. Selecting a patch in this ring contributes $(y - f_r/m_r)/k$ to enrichment, where y is its foreground indicator. Foreground contributions are nonnegative and background contributions are nonpositive. Selecting all k foreground patches therefore attains the maximum

$$E_{\max} = 1 - \frac{1}{k} \sum_{r:m_r>0} \frac{f_r^2}{m_r} \;\le\; 1 - \frac{k}{M}.$$

The inequality follows by Cauchy–Schwarz. It bounds the score itself. The possible change from R0 also depends on R0's starting score. A large raw improvement for small objects can thus reflect properties of both the map and the metric.

Area and centroid displacement. Object area is the share of valid patch slots labelled foreground. Centroid displacement is the distance from their centroid to the grid centre, in patches. For each checkpoint we computed the empirical quartiles on its finite-geometry images, assigning ties to the lower bin. The DeiT area cut points were 0.158, 0.342 and 0.571. DINO's different crop produced different masks. We paired scores within the same supplementary run and resampled images 2,000 times. In DeiT-S and DeiT-B the largest-minus-smallest area quartile differences in the C3 gain were -0.090 and -0.157 for enrichment. Comparing the most off-centre with the most central quartile instead gave $+0.057$ and $+0.095$. Area and centroid displacement covary. Among the smallest objects the latter differences were $+0.041$ [0.028, 0.053] and $+0.047$ [0.028, 0.066]. Some joint bins are small and these are observational contrasts.

Metric sensitivity. After seeing the enrichment gradients we examined patch AUROC under the same quartile cut points. AUROC is the probability that a foreground patch outranks a background patch, counting ties as one half. In the supervised models the gain in AUROC persists across area quartiles (Table E.2). AUROC does not condition on radius, so this comparison changes spatial adjustment as well as how foreground patches are selected. Overall C3-minus-R0 AUROC changes are $+0.028$, $+0.046$ and -0.005 for DeiT-S, DeiT-B and DINO.

Historical sample membership. Archived ImageNet-S files record zero enrichment for all-foreground grids, as the ring formula implies. Counts are 241 such grids for each DeiT and 265 for DINO. An additional 269/275 images, respectively, have zero object area and are excluded from the archived enrichment means. AUROC requires both classes, giving 11,909 DeiT and 11,879 DINO images. Restricting enrichment to that common eligibility changes C3-minus-R0 from 0.03757 to 0.03833 in DeiT-S, 0.06539 to 0.06671 in DeiT-B and -0.00351 to -0.00359 in DINO. The original verdicts use the archived cohort. In two DeiT-S images the original and supplementary R0 exports differ slightly. The stratified comparisons pair both methods from the supplementary export.

Table E.2: Mean C3 minus R0 by foreground-area quartile on ImageNet-S. Q1 contains the smallest objects. Both metrics use the same within-model cut points. AUROC excludes all-foreground crops.

Metric	Model	Q1	Q2	Q3	Q4
Enrichment	DeiT-S	0.092	0.041	0.013	0.002
Enrichment	DeiT-B	0.165	0.060	0.026	0.008
Enrichment	DINO	−0.003	−0.004	−0.005	−0.003
AUROC	DeiT-S	0.024	0.030	0.031	0.026
AUROC	DeiT-B	0.042	0.048	0.048	0.046
AUROC	DINO	0.000	−0.002	−0.006	−0.013

E.2 Independent pet-image controls

The Oxford release provides test filenames and trimaps. We sorted filenames by SHA-256 of the fixed prefix and name (protocol under `experiments/spatial/`), taking the first 384. Images were resized on the short side to 248 pixels with bicubic interpolation and centre-cropped to 224. The masks received the same spatial transform with nearest-neighbour interpolation. Label 3 (boundary) was ignored. A patch counted as foreground when at least half its valid pixels were foreground. One crop contained only foreground. Each image and its adjacent partner in hash order exchanged norm rankings. In each within-ring shuffle all heads shared a permutation for that image and layer. The recipient's sorted values, head scales and marginal distributions were preserved by rank transfer. Norm-only sets each head's received attention to its spatial mean before multiplying by its patch-specific length.

All seven readouts were evaluated on identical masks. Primary intervals used 10,000 paired image-bootstrap draws (seed 2026). A follow-up sensitivity check resampled the 192 donor pairs, weighting by the number of eligible images in each draw. C3 minus donor-rank gives 0.0223 [0.0187, 0.0261] in this check. The captured C3 map and logits were checked against the original implementation on the first batch. Each query–patch pair satisfied the triangle bound within numerical tolerance over the run. Public Oxford images and the model weights are available from their original providers. Individual scores, selected IDs, scripts and protocols accompany the source.

E.3 Last-block CLS value intervention

A second fixed filename hash selected 128 of the 384 images independently of their scores. At block 12 we cached its input and the unmodified logits. For each patch j we held the original CLS attention weights fixed and removed $t_j = \sum_h a^h_{\mathrm{CLS},j} z^h_j$ from the CLS attention output. The shared projection bias stayed in place. The remaining residual, token-wise MLP, final normalisation and classifier were evaluated for each modified CLS. If m is the margin between the original highest and second-highest logits, the response was $|m_{-j} - m|$ for those same classes. This intervention concerns the contextualised token at that slot. Its input evidence can span multiple image regions.

For each image, we correlated the 196 responses with each unmodified layer-12 readout by Spearman correlation. The means over images were 0.696 (all-query R0), 0.705 (all-query C3), 0.730 (CLS attention), 0.747 (CLS sum of head magnitudes) and 0.756 (CLS net magnitude). The paired CLS-only sum versus all-query C3 difference was 0.041 [0.026, 0.057] with 10,000 image-bootstrap resamples. Explicit interventions in full forward passes on four images and three patch indices each agreed with vectorised logits to within 2.4×10^{-6}. The 128-image run took about 3 seconds on the RTX 4060 with inputs cached. The supplied scripts and individual patch responses are under `experiments/cls/`.

Matched mask comparison. After inspecting the classifier intervention we evaluated the saved layer-12 rankings on the same 128 images and masks. All 128 had eligible foreground and background patches. Table 2 reports the three outcomes with equal image weights. Keeping the CLS query fixed, moving from the sum of per-head magnitudes to net magnitude changed margin Spearman by $+0.0092$ [0.0069, 0.0116], enrichment by $+0.0037$ [0.0007, 0.0069] and AUROC by $+0.0002$ [-0.0031, 0.0033]. The target margin uses the direct path queried by the CLS-specific score. These observations describe that layer and intervention. They do not identify the pixels responsible for earlier token representations.

F Key parts of the code

Given the attention a_{ij}^h and values v_j^h of a layer, the readings are:

```
def received_and_contribution(p, v, w_o, prefix=1):
    # p: [B,H,T,T]; v: [B,H,T,d]; w_o: [D,H,d]
    # T includes CLS; values include the value-projection bias.
    gram = torch.einsum('dhe,dhf->hef', w_o, w_o)
    length = torch.einsum('bhne,hef,bhnf->bhn', v, gram, v)
    length = length.clamp_min(0).sqrt()
    received = p.sum(2)  # sum over queries
    s = received.sum(1)[:, prefix:]
    c = (received * length).sum(1)[:, prefix:]
    return s, c

def relocate(s, probs):  # source j to predicted position p
    return torch.einsum('bj,bjp->bp', s, probs)
```

For the independent pet experiment, the following operations compare two weights for the same source patch. Here a has axes (image, head, query, patch), and z contains each head's projected value vector. The implementation processes queries in chunks.

```
# Own lengths versus the same lengths in a donor image's spatial order
length = z.norm(dim=-1)                          # [B, H, 196]
order = donor_length.argsort(dim=-1)
borrowed = torch.empty_like(length).scatter(-1, order, length.sort(-1).values)
received = a.sum(dim=2)
own_map = (received * length).sum(dim=1)
donor_map = (received * borrowed).sum(dim=1)

# Magnitude after summing the head vectors for each query and patch
net_map = sum(torch.einsum('bhij,bhjd->bijd', chunk, z).norm(dim=-1).sum(1)
              for chunk in a.split(32, dim=2))
```

For the final-block intervention we hold the attention matrix fixed. This excerpt from the full-forward check removes one patch's term from the CLS query. The output projection retains its bias and the rest of the model runs normally.

```
# p: [batch, head, query, source]; v: [batch, head, source, head_dim]
head_out = p @ v
head_out[:, :, 0, :] -= p[:, :, 0, source_patch + 1, None] * v[:, :, source_patch + 1, :]
full = head_out.transpose(1, 2).reshape(batch, tokens, -1)
return attn.proj_drop(attn.proj(attn.norm(full)))
```

AI Use Statement. I set the direction of this project, chose a fast-weight perspective for studying ViT attention maps, and organized the paper around three assumptions about how these maps are interpreted. I used AI tools to help search the literature, design follow-up experiments, write code, run experiments, analyze results, and draft and revise the report.